

## EducoWater Premium Wi-Fi

**Multi-Application Controller for Pools/Aquarium:**

( ) pH; ( ) ORP; ( ) Temperature; ( ) 4 Timers

**Hydroponics:**

( ) EC; ( ) TDS; ( ) Salinity; ( ) Temperature; ( ) 4 Timers

**Solar / HVAC / Thermal Systems:**

( ) Temperatures x2; ( )  $\Delta$  Differential Temperature;

( ) Average Temperature; ( ) 4 Timers

**Control 2:** ( ) digital temperature (DS18B20)

**Control 3:**

Conductivity Family: ( ) EC or ( ) TDS or ( ) Salinity or Temperature Family: ( ) Analog temperature or ( )  $\Delta$

**Differential temperature (Control 3 – Control 2) or ( )  $\Sigma$  average temperature (Control 3 + Control 2)/2**

**Control 4:** ( ) pH or ( ) ORP / Redox

Each control channel includes an independent **time modulation timer** ( ).

This allows cyclic ON/OFF modulation of the control output, useful for applications such as:

EducoWater includes **2 Pumps 60 ml/min** that can be dynamically assigned to each control. This flexible architecture allows the installer to freely associate physical outputs with the desired control logic.

EducoWater can be fully **configured, monitored, and operated locally** through the integrated **LCD display and 3-button keypad**, allowing complete standalone operation without requiring any external device.

Each pump is managed by an **independent timer**, with configurable **ON/OFF cycles and activation modes stored in non-volatile memory**, making the system ideal for **time-based, sensor-based, or hybrid dosing strategies**. EducoWater-Premium is **Smart Home compatible**, supporting **HTTP GET/SET commands via POST**, and can be easily integrated with **Node-RED, Home Assistant, custom scripts, or Alexa**.

For advanced connectivity, the system includes an integrated **Wi-Fi web server**, making the controller accessible from any smartphone, tablet, or PC using a standard web browser — **no app installation required**.

EducoWater supports multiple connection modes:

- Wi-Fi Access Point mode** for direct local connection (**no internet required**)
- Wi-Fi Station mode** for integration into your existing home or industrial network
- Remote worldwide access** through router port forwarding
- HTTP command interface** for smart home and automation integration
- The device also supports industrial automation integration through the integrated Modbus TCP Server interface, allowing communication with **PLCs, HMIs, and SCADA systems**.

The integrated web platform provides a modern and convenient interface for:

- live monitoring of all control channels
- sensor calibration
- control configuration
- timer management
- relay assignment
- Wi-Fi configuration
- remote diagnostics

Advanced intelligent cloud-connected services are also available:

### NetSky AI Copilot

AI-assisted support for:

- troubleshooting technical problems
- answering product and manual-related questions
- guided diagnostics
- assisted configuration suggestions
- automation command generation

### Sentinel Monitoring

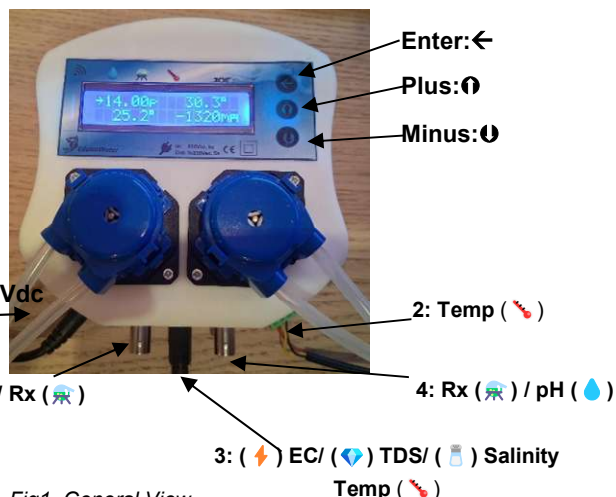


Fig1. General View

### Product Description:

EducoWater is a multifunction Wi-Fi controller designed for water treatment, hydroponics, aquariums, swimming pools, solar thermal systems, HVAC applications, and general process automation.

The system combines multiple measurement technologies and flexible control logic in a single Wall Mounting device.

EducoWater supports **5 different sensor technologies**:

- ( ) pH
- ( ) ORP / Redox
- ( ) Digital temperature sensors (DS18B20)
- ( ) Analog NTC temperature
- ( ) EC / ( ) TDS / ( ) Salinity

Advanced ( ) **thermal operating modes** allow intelligent multi-point temperature control for demanding applications such as solar thermal systems, HVAC automation, circulation pumps, heat exchangers, fireplaces with water jackets, and general industrial thermal processes.

Available modes include:

- Standard temperature control** (single or dual sensor based)
- Dual temperature monitoring**
- Differential temperature control** (ideal for solar collector / tank  $\Delta T$  applications)
- Average temperature control** (combined thermal regulation based on the mean value of two measurement points)

The controller provides **4 fully independent control channels**, each configurable according to the selected application. The following architecture is available:

**Control 1:** ( ) pH or ( ) ORP / Redox

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Advanced alarm supervision with:

- minimum / maximum threshold monitoring
- persistence filtering to avoid false alarms
- automatic alarm notifications
- recovery notifications
- email alert messaging

#### Reporter Data Logging

Integrated reporting and historical monitoring with:

- trend visualization
- scheduled email reporting with Excell data

The controller is designed for **Wall Mounting** and powered directly from **Power Supply 12Vdc**.

#### Installation application example:

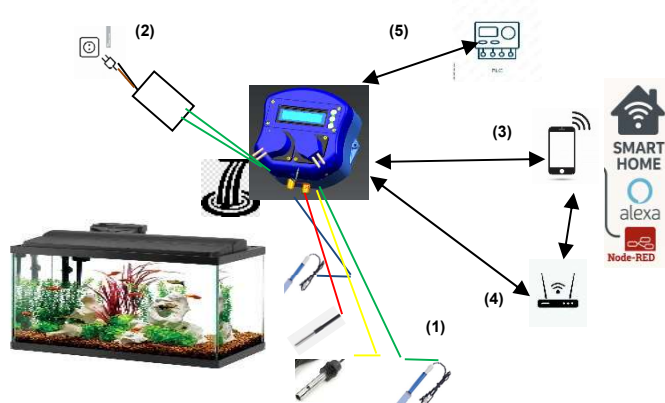


Fig.2 Typical Installation

#### 1. Connect the Sensors

Connect the sensors according to the specific application:

-BNC Connector: pH or ORP / Redox probe

-Terminal Block Connector: DS18B20 digital temperature sensor

-Terminal Block Connector: EC probe or Analog temperature sensor

-BNC Connector: pH or ORP / Redox probe

#### 2. Power Connection and Outputs

Connect the external power supply 230Vac/12Vdc

#### 3. Connect a Mobile Device

Use your smartphone or tablet to access the device's interface. This can be done by connecting directly to the device's local hotspot or through your home Wi-Fi network.

#### 4. Connect to Wi-Fi (Optional)

For remote monitoring and control, connect the device to your Wi-Fi network. Follow the on-screen instructions in the mobile interface to configure your network credentials.

#### Normal Function (use with Display and buttons):

When the controller is powered on, it loads the control data and the message 'Loading...' appears on the display. After a

few seconds, the main screen appears, showing the status of the four controls, as illustrated in Fig. 2.

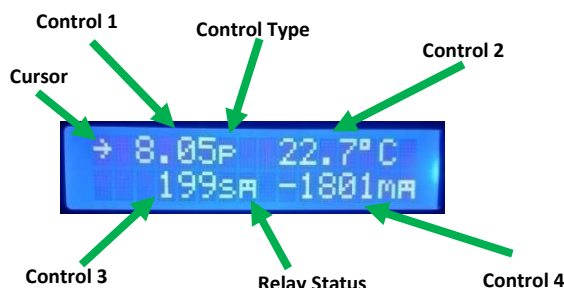


Fig3. Normal Function

On the main screen, the following elements are displayed:

- **Cursor**,

- **Controls**: current measured values for each control,

Control 1: pH / Redox,  
Control 2: Digital Temperature Sensor  
Control 3:  
 EC/() TDS/() Salinity/  
 Analog/ Differential/ Average Temperature  
Control 4: pH / Redox,

- **Control type** indicator that can be:

- "p" for () PH
- "m" for () Redox in mv,
- "°" for Temperature () in °C
- "°" (lower) for Average Temperature () in °C
- "Σ" for Differential Temperature () in °C
- "c" for () EC in us/cm
- "d" for () TDS in ppm
- "s" for () Salinity in ppm

**Note:** if the timer is displayed then the symbol "h" stays for hours "m" stays for minutes

- **Relay status** symbol that is blinking when the relay is on.

To move the cursor, press the buttons (⬅) and (➡) to shift the focus to the desired control.

To change the values and settings of a control, briefly press and release the **MODE** (⬅) button to enter the "Change Setting mode", as shown on the following page.

Pushing the button (⬅) for more than 10 seconds the cursor (➡) will start to blink and will be possible to modify the set up values with the buttons (⬅) and (➡). When the desired set up value is reached it will be stored in the controller pushing the button (⬅).

Following the procedure above is possible to change each set up value for each control function

Following the procedure above it is possible to change each set-up value for every control function.

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## Change Settings mode

The device configuration menu is structured as shown in the figure below. Each branch corresponds to one control channel.

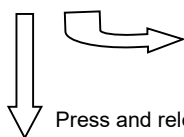
To access the settings of a specific control, move the cursor to the desired channel using the **(N)** and **(U)** buttons, then press and release the **MODE (←)** button. The controller enters the Control Settings Mode for the selected channel, and the cursor starts flashing to indicate that the setting editing mode is active.

### Show Actual Values Page:

Cursor: Fix

Value: Fix

Relay Symbol: Blinks if relay is on



Press **(N)** and **(U)** buttons to move the cursor to next control.

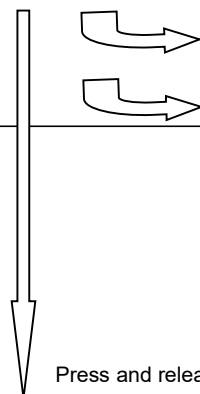
Press and release **MODE (←)** button to access the "Show  Set Point value Page"

### Show Set Point value Page:

Cursor: Fix

Value: Blinks

Symbol: Blinks



Press **(N)** and **(U)** buttons to  Return to Show Actual Values Page.

Press **MODE (←)** button until u enter in "Change  Set Point value Page"

### Change Set Point value Page:

Cursor: Blinks

Value: Fix

Symbol: Fix



- Press **(N)** and **(U)** buttons to change the value.
- Press **MODE (←)**, button until save the value and  Return to Show Actual Values Page.

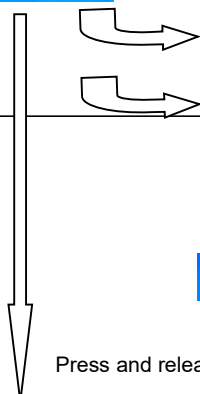
Press and release **MODE (←)**, button to access the "Show  Hysteresis value Page"

### Show Hysteresis value Page:

Cursor: Fix

Value: Blinks

Symbol: Blinks



Press **(N)** and **(U)** buttons to  Return to Show Actual Values Page.

Press **MODE (←)** button until u enter in "Change  Hysteresis value Page"

### Change Hysteresis value Page:

Cursor: Blinks

Value: Fix

Symbol: Fix



- Press **(N)** and **(U)** buttons to change the value.
- Press **MODE (←)** button until save the value and  Return to Show Actual Values Page.

Press and release **MODE (←)** button to access the "Show  Logic value Page"

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### Show Logic value Page:

Cursor: Fix

Value: "Pos" for positive or "Neg" for negative alternatig with the placeholder "Logic"

Symbol: none

→Logic → Pos

Press (⏏) and (⏏) buttons to [Return to Show Actual Values Page](#).

Press MODE (←) button untill u enter in "Change  Logic value Page"

#### Change Logic value Page:

Cursor: Blinks

Value: Fix

Symbol: none

Pos

-Press (⏏) and (⏏) buttons to change the value.

- Press MODE (←),button untill save the value and [Return to Show Actual Values Page](#).

Press and release MODE (←) button to access the "Show  Timer Status Page"

### Show Timer Status Page:

Cursor: Fix

Value: Fix

Symbol: Blinks if Timer is in "On Phase"

→18:13m

Press (⏏) and (⏏) buttons to [Return to Show Actual Values Page](#).

Press and release MODE (←) button to access the "Show  Timer Active Status Page"

### Show Timer Active Status Page:

Cursor: Fix

Value: "Active" for active or "NotAct" for not active alternatig with the placeholder "Timer"

Symbol: none

→Timer →Active

Press (⏏) and (⏏) buttons to [Return to Show Actual Values Page](#).

Press MODE (←) button untill u enter in "Change  Timer Active Status Page"

#### Change Timer Active Status Page:

Cursor: Blinks

Value: Fix

Symbol: none

NotAct

-Press (⏏) and (⏏) buttons to change the value.

- Press MODE (←),button untill save the value and [Return to Show Actual Values Page](#).

Press and release MODE (←) button to access the "Show  Timer On value Page"

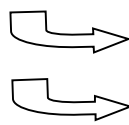
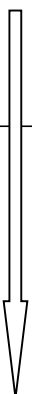
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### Show Timer On value Page:

Cursor: Fix

Value: Blinks

Symbol: Blinks



Press (N) and (U) buttons to  Return to Show Actual Values Page.



Press **MODE (←)** button untill u enter in "Change  Timer On value Page"

### Change Timer On value Page:

Cursor: Blinks

Value: Fix

Symbol: Fix



- Press (N) and (U) buttons to change the value.
- Press **MODE (←)**,button untill save the value and  Return to Show Actual Values Page.

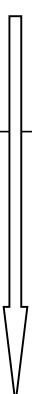
Press and release **MODE (←)**,button to access the "Show  Timer Off value Page"

### Show Timer Off value Page:

Cursor: Fix

Value: Blinks

Symbol: Blinks



Press (N) and (U) buttons to  Return to Show Actual Values Page.



Press **MODE (←)** button untill u enter in "Change  Timer Off value Page"

### Change Timer Off value Page:

Cursor: Blinks

Value: Fix

Symbol: Fix



- Press (N) and (U) buttons to change the value.
- Press **MODE (←)**,button untill save the value and  Return to Show Actual Values Page.

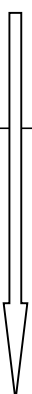
Press and release **MODE (←)**,button to access the "Show  Sensor Calibration point Low value Page" -> (not applicable for Temperarue controls)

### Show Sensor Calibration point Low value Page:

Cursor: Fix

Value: Blinks

Symbol: Blinks



Press (N) and (U) buttons to  Return to Show Actual Values Page.



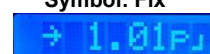
Press **MODE (←)** button untill u enter in "Change  Sensor Calibration point Low value Page"

### Change Sensor Calibration point Low value Page:

Cursor: Blinks

Value: Fix

Symbol: Fix



- Press (N) and (U) buttons to change the value.
- Press **MODE (←)**,button untill save the value and  Return to Show Actual Values Page.

Press and release **MODE (←)**,button to access the "Show  Sensor Calibration point High value Page"

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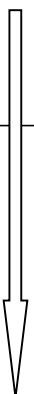
### Show Sensor Calibration point High value Page:

Cursor: Fix

Value: Blinks

Symbol: Blinks

→ 12.45P



Press (F1) and (F2) buttons to  Return to Show Actual Values Page.



Press MODE (←) button until u enter in "Change  Sensor Calibration point High value Page"

### Change Sensor Calibration point High value Page:

Cursor: Blinks

Value: Fix

Symbol: Fix

→ 12.42P



- Press (F1) and (F2) buttons to change the value.
- Press MODE (←),button until save the value and  Return to Show Actual Values Page.

Press and release MODE (←) button to access the "Show  Relay Associated Page"

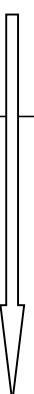
### Show Relay Associated Page:

Cursor: Fix

Value: Blinking either "NoRel" for no relay associated or "Rel1 to Rel4" for relay 1 to 3 if associated to current control

Symbol: Relay

→ NoRel



Press (F1) and (F2) buttons to  Return to Show Actual Values Page.



Press MODE (←) button until u enter in "Change  Relay Associated Page"

### Change Relay Associated Page:

Cursor: Blinks

Value: Fix

Symbol: Relay

Rel1



- Press (F1) and (F2) buttons to change the value.
- Press MODE (←),button until save the value and  Return to Show Actual Values Page.

Press and release MODE (←) button to access the "Show  Type Page"

### Show Type Page:

Cursor: Fix

Value: Blinking either "PH"/"Redox"/"EC"/"TDS"/"SAL" for (  ) PH, (  ) Redox, (  ) EC, (  ) TDS, (  ) Salinity types.

Symbol: none

→ Ph



Press (F1) and (F2) buttons to  Return to Show Actual Values Page.



Press MODE (←) button until u enter in "Change  Type Page"

### Change Type Page:

Cursor: Blinks

Value: Fix either "PH"/"ORP"/"EC"/"TDS"/"SAL" for (  ) PH/(  ) Redox/(  ) EC/(  ) TDS/(  ) Salinity types.

Symbol: none

→ Redox



- Press (F1) and (F2) buttons to change the value.
- Press MODE (←),button until save the value and  Return to Show Actual Values Page.

Press and release MODE (←) button to  Return to Show Actual Values Page.

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### Calibration Notes

-For best calibration accuracy, choose two calibration values that are as far apart as possible.  
 -Prepare the calibration solutions before starting the procedure (for example, pH 1.00 and pH 10.01).  
 -Place the probe in the first calibration solution and wait a few seconds until the reading becomes stable. Only then start the calibration by pressing the (←) button as explained above.

-If an error occurs during calibration, the default calibration values can be restored from the Calibration Procedure window by pressing the ← → U buttons simultaneously.

### Power Input:

| Number | Label | Description    |
|--------|-------|----------------|
| 8      | L     | Neutral 230Vac |
| 9      | N     | Line 230Vac    |



Fig4a.. Power Input 12Vdc

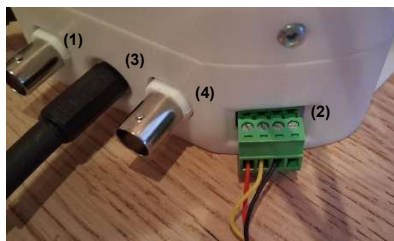


Fig4b.. Sensors Interface

### Sensors Wiring:

| Label               | Description                 |
|---------------------|-----------------------------|
| (1)Ph/Orp           | Sensor 1 (BNC connector)    |
| (2)Temperature      | Sensor 2 (DS18B sensor)     |
| (3)EC/TDS/SALT/Temp | Sensor 3 (EC sensor) or NTC |
| (4)Orp/Ph           | Sensor 4 (BNC connector)    |

### Channel 2: Temperature Pin Out

| Color  | Description   |
|--------|---------------|
| Red    | V+            |
| Yellow | Digital Input |
| Black  | GND           |
| N/A    | N/A           |

### Wi-Fi Connection: Direct Mode.

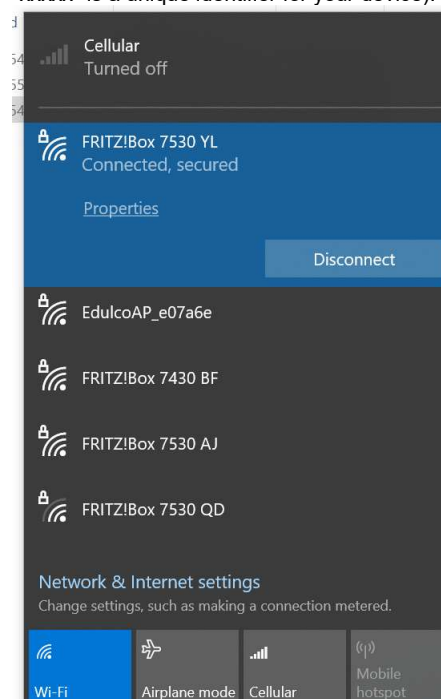
The system can connect to any device that supports Wi-Fi connectivity, such as smartphones, tablets, or PCs.

The device creates its own Wi-Fi access point. Simply connect your device to the EducoWater network to access the interface without requiring an internet connection.

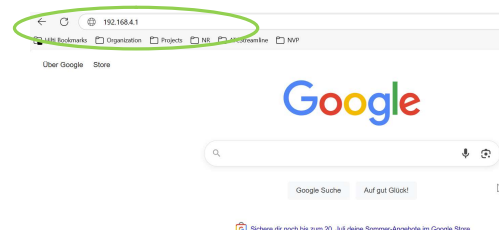


Fig.5 Direct mode WiFi connection

-Open the Wi-Fi settings on your smartphone, tablet, or PC and look for a Wi-Fi network named **EducoAP\_XXXXX** (the "XXXXX" is a unique identifier for your device).



-Select the network and connect — no internet is required.  
 -Once connected, open your web browser (for an example Internet Explorer or Google) and enter the device's local IP address **192.168.4.1** to access the control interface.  
 -Default Password: **"123456789"**



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## Studios Web Page Interface



Fig.6 Web Page Interface

This is the main page that opens by default when accessing the device.

At the top (1), the navigation bar provides quick access to the following pages:

- **Monitor** – Shows system status.
- **Sensors** – Sensor type definition and Calibration.
- **Controls** – Configure logic hysteresis and setpoints.
- **Timers** – configure timers functions
- **Relays** – hw and remote relays configuration
- **WiConfig** – Manage Wi-Fi setup (Access Point or Station mode), Email contact for smart and AI functions
- **Sentinel** – Safety monitoring with min/max thresholds, persistence time and alarm detection
- **Reporter** – Automatic status and alarm reports via email, with configurable schedules
- **NetSky** – AI-assisted diagnostics, smart suggestions and system interaction
- **About** – Displays system information and useful links.

## Monitor Page

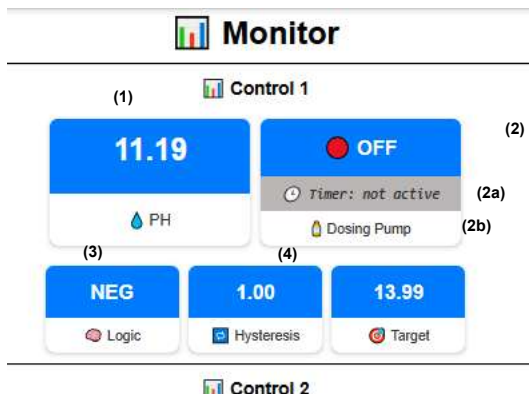


Fig.7 Monitor Page

This page displays the current status of the device, including:

- (1)Current sensor reading (pH or Redox)
- (2)Current state of the actuator (ON/OFF)

-(2a) Actuator Status

-(2b) Timer status

Note: The actuator will be ON **only** if the control logic is active (e.g., pH below the threshold in positive logic) **and** the timer is either disabled or currently in its ON phase.

This allows **time-based modulation** of the actuator using the timer function, even when the control condition remains continuously true.

(3)Regulation logic (Negative: actuator tries to reduce the current value till the set point; Positive: actuator tries to increase the current value till the set point)

(4)Hysteresis (range of no action across the set point)

(5)Set point setpoint

The same logic and conditions apply also to **Control 2, 3, 4**.

## Sensors



Fig.8 Calibration Page

The Calibrator section is used to calibrate the connected sensor (pH or Redox). It consists of two main elements:

- (1) Sensor Type Setting
- Allows selecting the type of sensor to calibrate (pH or Redox).

- (2) Calibration Input Area

Used to enter the reference values of the calibration solutions for each calibration points (Low ▼ or High ▲ ).

- (3) Message Box

Displays:

the current sensor type and live measured value, and

the currently stored calibration points (Low ▼ or High ▲ ).

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### How to Perform a Calibration

To carry out the calibration, prepare two standard buffer solutions. Then press Enter to begin. The message box will guide you through the following steps:

- I. Insert the sensor into the first buffer solution, then press Enter.
  - II. Enter the known value of the first solution and press Enter.
  - III. Wait approximately 10 seconds for the reading to stabilize, then press Enter.
  - IV. Insert the sensor into the second buffer solution and press Enter.
  - V. Enter the known value of the second solution and press Enter after waiting around 10 seconds for stabilization.
- Once all steps are completed, the calibration will be successfully saved and applied.

#### Note:

- To cancel the calibration process at any time, press the **Restart** button.
- To restore the factory calibration parameters, press the **Reset** button.

#### Note:

Using **only a single calibration point** may introduce **measurement errors**, as the system cannot fully compensate the sensor slope. For this reason, it is **strongly recommended** to always perform a **two-point calibration**, selecting reference values as far apart as possible, in order to achieve the highest accuracy and stability.

The same logic and conditions apply also to **Sensor 4** (2 can be only temperature and doesn't need calibration, 3 is at the moment used only as Timer).

### Controls

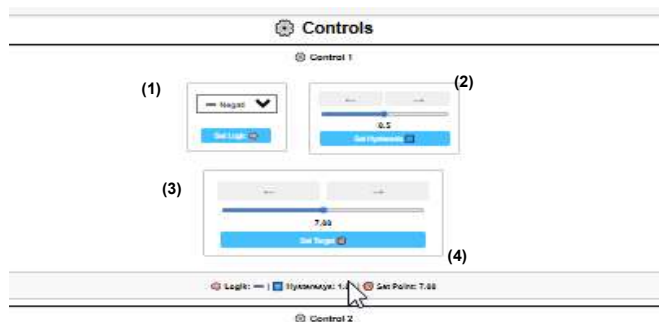


Fig.7 Settings Page

This page allows the user to configure the control behavior through the following sections:

- (1) Logic Setting Controls  
Define whether the actuator works in Negative (tries to reduce the value) or Positive logic (tries to increase the value).
  - (2) Hysteresis Setting Controls  
Set the hysteresis range around the set point where no action is taken, avoiding rapid ON/OFF toggling.
  - (3) Target Setting Controls  
Define the desired set point for the sensor (e.g., target pH or Redox value).
  - (4) The selected settings are immediately reflected in the status panel below, as shown in the figure.
- By pushing the buttons in each section, the configuration can be adjusted in real time.

The same logic and conditions apply also to **Control 2, 3, 4**.

### Timers

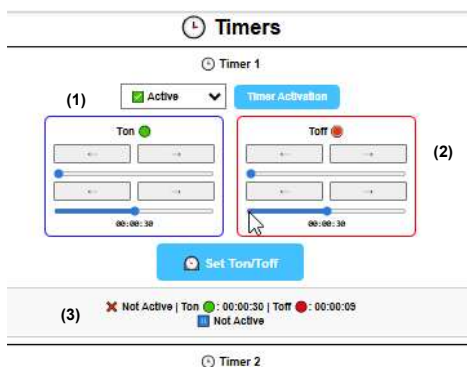


Fig.8 Timers Page

In the **Timers Page**, it is possible to configure the timer-based control of the actuator:

#### (1) Enable or disable

To activate or deactivate the timer, use the dropdown menu at the top, and press the **"Timer Activation"** button to apply the selected state.

(2) Set the **TON** time (the ON duration) using the scroll bars and adjustment buttons.

(3) Set the **TOFF** time (the OFF duration) in the same way. After adjusting the values, press the **"Set TON/TOFF"** button to apply the configuration.

(4) The **Status Panel** at the bottom of the page provides a clear summary:

-(4a) The first row shows the current timer settings: whether it is **Active** or **Not Active**, and the configured **TON** and **TOFF** times.

-(4b) The second row displays the **current state of the relay** (ON or OFF), based on the timer cycle and control condition.

➔ The same configuration logic applies to **Timer 2**, which is associated with **Control 2, 3, 4**.

### Relays



Fig.12 Outputs Page

(1) In this section you can select which physical relay on the board is assigned to each control.

By selecting the forbidden symbol (⊘), the hardware relay is released from any control and becomes available for other assignments.

(2) Hardware relay association status panel.

The same applies to **Hw Relay 2, 3, 4**.

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**WiFi Configurator**

**(1) Access Point**

SSID:  Pass:

**(2)** SSID: EducoAP-14B0 IP: 192.168.4.1

**(a) Station** (b) SSID:  Pass:

**(c)** SSID: FRITZBox 7530 YL IP: 192.168.178.144

**Email Contacts**

(I) Receiver:  Save Clean

(II) Sender:  Pass:

(III) Receiver: tablanodag@hotmail.com Sender: edulcowater.mailer@gmail.com (Limited)

Fig. 10 WiFi configuration Page

The WiFi Configurator allows the device to connect to a private WiFi network for access via a local router or even through the internet worldwide.

#### (1) WiFi Credentials Input

In this section, you can enter the credentials (SSID and Password) of your private WiFi network.

By pressing the Save button, the device will attempt to connect to the router using the provided information.

#### (2) Access Point Status and Information

This section displays information related to the device's Access Point mode. When the device operates as an Access Point, it shows the SSID (network name) being broadcast, the IP address to be used for connection, and other relevant connection parameters. This allows direct configuration without requiring an external WiFi network.

#### (3) Station Mode Configuration and Status

This section allows you to save the credentials (SSID and Password) of your private WiFi network (router). Once saved, the device will attempt to connect to the specified network. The panel displays the current connection status. When successfully connected, the assigned IP address will be shown.

At this point, the web interface can be accessed from any device connected to the same WiFi network by entering the shown IP address in a browser.

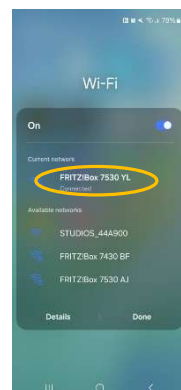
**By pressing the clean button (🗑️) is possible to cancel the stored (🔑) credentials.**

To connect the device to a local WiFi router:

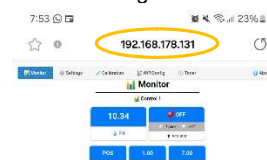
(a)-Enter the **SSID** and **Password** of your private WiFi network in the designated fields.

(b)-Press the **Save** button to initiate the connection process. Once connected, the system will automatically obtain an **IP address** from the router (c). This IP address will be displayed in the **message section**.

After the connection is established, any device (such as a smartphone, tablet, or PC) connected to the **same WiFi network** can access the **Studios Web Page Interface** using a standard web browser.



Simply enter the assigned IP address into the browser's address bar, as shown in the example. Multiple devices can connect to and interact with the StudiosManager interface **simultaneously**.



#### "Email Contacts"

The WiFi Configurator allows the device to connect to a private WiFi network for access via a local router or even through the internet worldwide.

The WiFi Configurator allows the device to connect to a private WiFi network for access via a local router or worldwide through the internet.

(I) In this section you can define the **recipient email address** used by all advanced functions such as **Sentinel**, **NetSky**, and **Reporter**, which are described in the following sections.

(II) Here you can configure the **sender email account (SMTP/IMAP service)** used to send notifications.

By default, the system is preconfigured with the **edulcowater.mailer@gmail.com**, which is limited to **5 emails**. For **privacy reasons** and **unlimited usage**, it is recommended to configure your **own email account** by following the guide below in the next "**Note**" chapter.

(III) The **status panel** reflects the current state of the **email configuration and settings**.

**NOTE:** How to create a private gmail sender email

#### 📧 Email Configuration (Gmail Setup)

To allow your EducoWater device to send alert messages via Gmail, you need to create a dedicated Gmail account and enable a special **App Password**.

Follow these steps carefully:

##### 1 Create a Gmail account

It is recommended to use a dedicated account (for example: **edulcowater.alerts@gmail.com**) instead of your personal one.

##### 2 Enable 2-Step Verification

-Go to <https://myaccount.google.com/security>

-In the section "**Signing in to Google**", click **2-Step Verification**

-Follow the steps to activate it (you'll need your phone).

##### 3 Generate an App Password

After 2-Step Verification is enabled:

-Return to the same security page

-Click **App passwords** (or go directly to

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<https://myaccount.google.com/apppasswords>

- Sign in if requested
- Under **Select App**, choose **Mail**
- Under **Select Device**, choose **Other (Custom name)** and type: "MyEducoWaterSender"
- Click **Generate**

#### 4 Copy the generated password

A yellow window will appear with a 16-character password, for example:

"abcd efgh ijkl mnop"

Copy it **without spaces** → "abcdefghijklmnop"

#### 5 Enter your credentials in the EducoWater web interface

- Open the configuration page
- In the field **Sender**, type your Gmail address (e.g. edulcowater.alerts@gmail.com)
- In the field **Pass**, paste the 16-character App Password (without spaces)
- Press **Save**

#### 6 Restart or reconnect the Email service

The device will store your credentials in memory and automatically reconnect to Gmail's SMTP server.

#### Important Notes

- Regular Gmail passwords **do not work**. You must use the App Password.
- Do **not share** the App Password with others.
- If you change or delete it from your Google account, generate a new one and re-enter it in the device.
- Each App Password can be revoked anytime from your Google Account → **Security** → **App Passwords**.

#### Remote Access (Worldwide):

If **port forwarding** is configured on your router, the system can also be accessed **remotely from anywhere in the world**. Please consult your internet service provider or router documentation for instructions on how to enable and configure port forwarding.

As an example u can find one explaining link for the FritzBox router

<https://www.youtube.com/watch?v=uSC1-iWqibE>

In the following picture there is an illustration of all the possible ways to access the device.

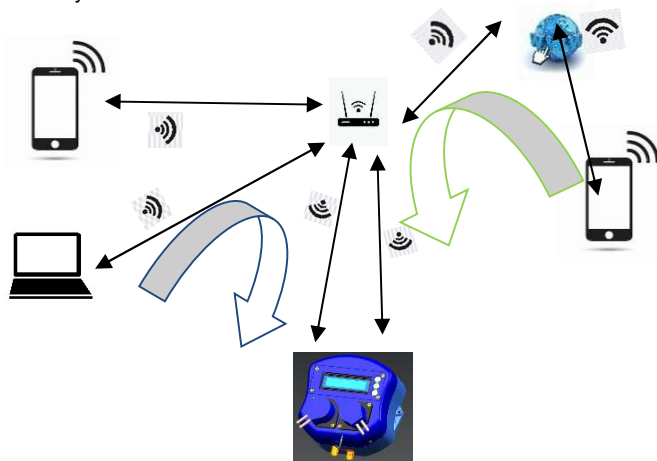


Fig.11 Remote mode WiFi connection

#### Sentinel

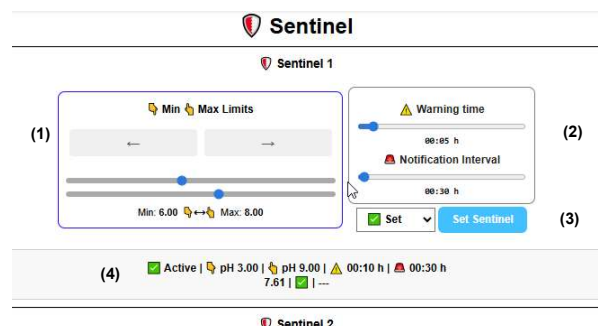


Fig.12 Sentinel Page

The Sentinel function continuously monitors a control value and sends alert emails to the address configured in the WiFi Configurator whenever the value goes out of the allowed limits. It is designed to detect abnormal conditions and notify you reliably, avoiding false alarms.

#### (1) Limits configuration

Using the up and down arrows, you can set the minimum and maximum allowed values.

If the measured value goes outside these limits, the condition is considered out of range.

#### (2) Timing configuration

Warning Time: the time that the value must remain out of limits before it is considered a real alarm (filters short spikes).

Notification Interval: how often the alarm email is repeated if the alarm condition persists.

#### (3) Activation and save

With the activation button, the Sentinel can be set ACTIVE or NOT ACTIVE.

When active, the Sentinel starts monitoring using the configured limits and timings.

#### (4) Status panel

The status panel reflects the current Sentinel configuration and state, showing whether it is active and how it is operating.

If the monitored value **returns within the configured limits**, the Sentinel automatically sends a **recovery email** to inform that the alarm condition has been resolved.

This guarantees a **complete notification cycle**: alarm raised → alarm cleared.

Below one example of Alarm notification and alarm resolved notification

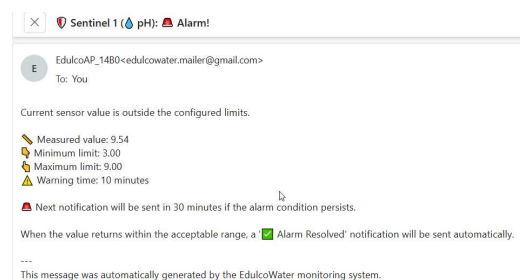


Fig.12a Sentinel Alarm notification email

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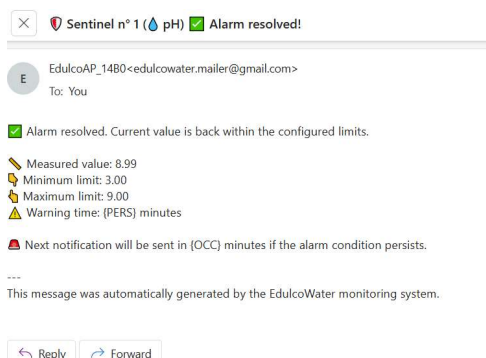


Fig.12b Sentinel Alarm resolved notification email

The same applies to Sentinels 2, 3, 4.

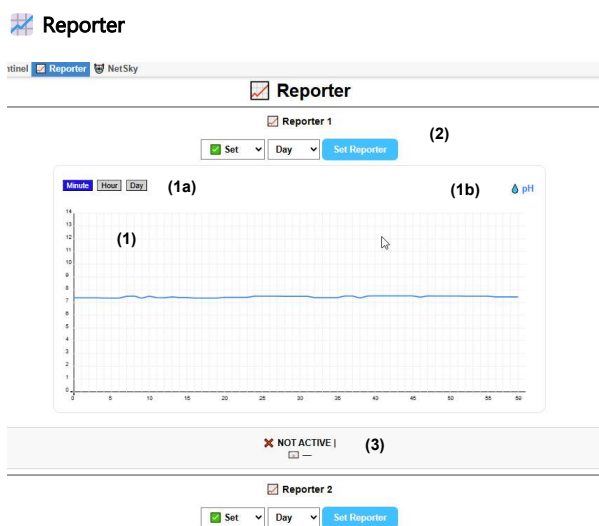


Fig. 13 Reporter Page

In this page you can monitor, in real time, the trend of the selected control value (pH, Redox, Temperature, etc.) and review its history over the last minute, hour, or day. You can also configure the Reporter to send periodic reports by email, according to your selected settings (recipient email is defined in the WiFi Configurator).

(1) Trend Graph

Here you can view the recorded values of the control.

(1a) Display range selector: choose the visualization interval (minute / hour / day).

(1b) Legend: indicates the control type and what is being plotted.

(2) Report Settings

Select the desired report type and configure whether you want the report to be sent according to your chosen schedule/settings.

(3) Status Panel

The status panel reflects the current reporting state, including the data collection progress (%) before the next report is sent to the email configured in the WiFi Configurator.

In the following pictures an example of the reporting email and of the reported data in excell file format

Fig.13a Reporter Email notification

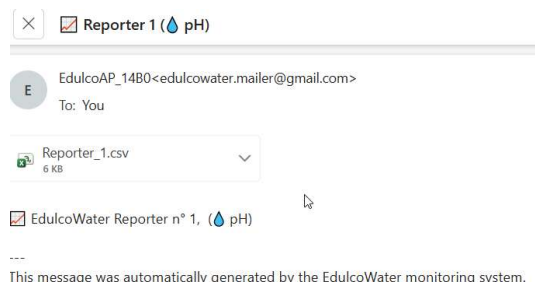


Fig.13b Excell data reported example

| # | TYPE: PH         | UNIT: pH |        |        |
|---|------------------|----------|--------|--------|
| 2 | Timestamp        | Average  | ValMin | ValMax |
| 3 | 12/13/2025 15:21 | 777      | 777    | 777    |
| 4 | 12/14/2025 15:21 | 777      | 777    | 777    |
| 5 | 12/15/2025 15:21 | 777      | 777    | 777    |
| 6 | 12/16/2025 15:21 | 774      | 774    | 774    |
| 7 | 12/17/2025 15:21 | 774      | 774    | 774    |
| 8 | 12/18/2025 15:21 | 777      | 777    | 777    |
| 9 | 12/19/2025 15:21 | 775      | 774    | 777    |
| 0 | 12/20/2025 15:21 | 774      | 774    | 774    |
| 1 | 12/21/2025 15:21 | 775      | 774    | 776    |
| 2 | 12/22/2025 15:21 | 776      | 776    | 776    |
| 3 | 12/23/2025 15:21 | 776      | 776    | 777    |
| 4 | 12/24/2025 15:21 | 777      | 777    | 777    |
| 5 | 12/25/2025 15:21 | 777      | 777    | 777    |
| 6 | 12/26/2025 15:21 | 777      | 777    | 777    |
| 7 | 12/27/2025 15:21 | 777      | 777    | 777    |
| 8 | 12/28/2025 15:21 | 777      | 777    | 777    |
| 9 | 12/29/2025 15:21 | 777      | 777    | 777    |
| 0 | 12/30/2025 15:21 | 777      | 777    | 777    |
| 1 | 12/31/2025 15:21 | 777      | 777    | 777    |
| 2 | 1/1/2026 15:21   | 777      | 777    | 777    |
| 3 | 1/2/2026 15:21   | 777      | 777    | 777    |
| 4 | 1/3/2026 15:21   | 777      | 777    | 777    |
| 5 | 1/4/2026 15:21   | 777      | 777    | 777    |
| 6 | 1/5/2026 15:21   | 777      | 777    | 777    |
| 7 | 1/6/2026 15:21   | 776      | 776    | 777    |
| 8 | 1/7/2026 15:21   | 776      | 776    | 776    |
| 9 | 1/8/2026 15:21   | 775      | 775    | 776    |
| 0 | 1/9/2026 15:21   | 775      | 775    | 775    |

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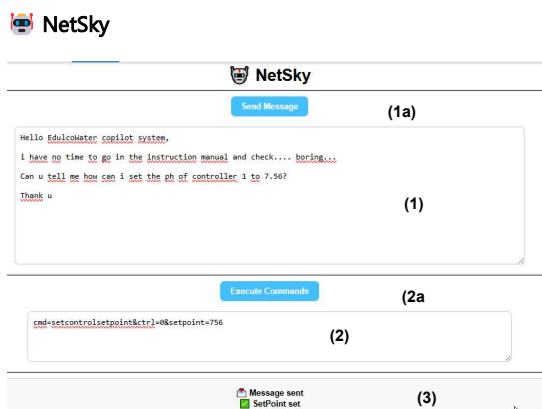


Fig.14 NetSky Page

NetSky is the first system on the market to integrate an AI Copilot assistant directly into the device.

By writing a message as shown in the example, you can receive intelligent assistance for diagnostics, user manuals, and other device-related information.

(1) Email Message

(1) Email message  
In this field you can write your request or description of the issue.

The message will be analyzed by the Copilot system to provide targeted assistance.

(1a) Send Button

Press the button to send the message by email and receive support from the NetSky Copilot system.

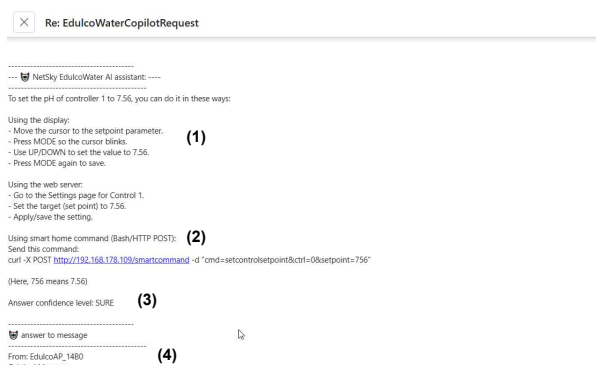
(2) Prompt commands

In this field you can copy and paste commands the NetSKY system suggests to u.

(2a) Commands Execution Button

**(2a) Commands Execution Button**  
Press the button to set the copied and pasted commands in order to adjust the EdulcoWater according to your needs.

*Fig.14a NetSky Email response*



In the reply email, the following sections are included:

(1) Explanation

Explanatory text with direct reference to the commands shown on the display and their meaning.

## (2) Bash / Control Commands

Examples of executable commands (e.g. for Smart Home integration) that can be copied and used directly.

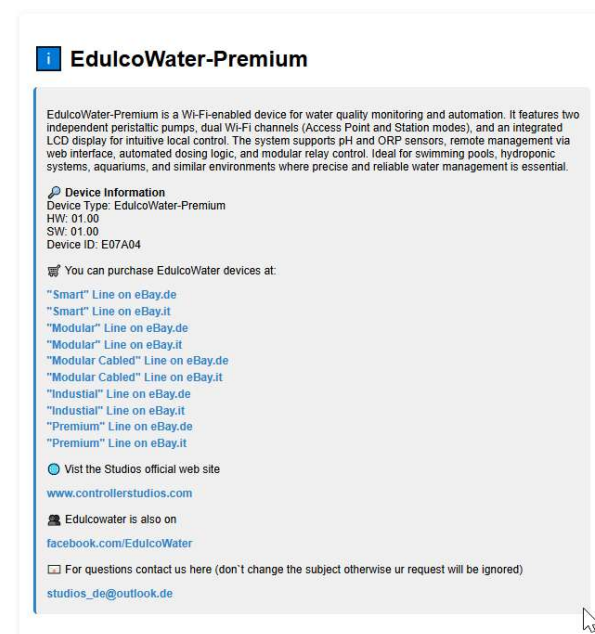
### (3) Confidence Level

An indication of the confidence level of the response, meaning how reliable the system considers the proposed solution (yeah... it's still growing 🤔).

#### (4) Request Details

Information related to the original request message, for traceability and context.

**About**



*Fig.15 About Page*

The About section provides a brief description of the product and useful resources for further support or information.

📄 Product Overview 🌐 Useful Links like e commerce company web site and 📧 email for technical support

### Technical Data:

| Technical Data:      |                                     |
|----------------------|-------------------------------------|
| Description          | Rating                              |
| Supply               | 12Vdc                               |
| Power consumption    | 12W                                 |
| Ph range             | 0.01-14.00                          |
| Redox range          | -1800/1800 mV                       |
| Control algorithm    | ON-OFF, Adjustable Hyster and Logic |
| Ph error             | ±0,01                               |
| Ph resolution        | ±0,01                               |
| Redox error          | ±1 mV                               |
| Temp. range          | 5°C-70°C                            |
| Temp. error          | + -1°C                              |
| Temp. resolution     | 0.1°C                               |
| Seonsor refresh time | 2 seconds                           |
| Ton-Toff range       | 1 sec -24 hours;                    |
| Max timer error      | + -10sec/day                        |
| Power regulation     | 33%-100%                            |
| Pump max flow        | 1100ml/h                            |
| WiFi Range (1)(2)    | 10mt                                |

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(1) The range can be extended in **Access Point mode** by using any standard WiFi repeater, and in **Station mode** through any router, whether connected to the internet or not.

(2) This device supports WPA/WPA2. It does not support WPA3. When using routers or mesh systems with mixed WPA2/WPA3 mode, connection issues may occur. In such cases, configure the access point to **WPA2 only** or use a repeater that supports WPA2.

**Mechanical Overall Dimensions:** 140x200x130 mm

**Markings:** CE

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## SMART HOME FUNCTIONS

### Using Smart Home Commands

The Smart Home commands of the EducoWater Premium system are designed to enable seamless integration with external platforms such as Amazon Alexa, Node-RED, Home Assistant, or any custom automation system that supports HTTP POST requests. These commands allow full remote access to the control logic, timers, calibration functions, and system configuration.

In the example shown below, a command is executed using a bash terminal to interact with the device through its local IP address.

See the figure below for a real example of sending a command from Bash (Linux/macOS/Windows terminal with curl).

To run a command from the terminal:

Open a terminal or command prompt on your PC or smart device.

Type the provided curl command as shown in the examples.

Replace the IP address with your device's actual IP (in the example it was **192.168.178.130** but u can get your own in the WiFiConfiguration page panel statu).

Press ENTER to send the command.

MINGW64:/c/Fabiano/05\_Projects/03\_Studios/53\_Edulcowater-Smart/70\_Manuals

```
DAQUFAB@K123716H1 MINGW64 /c/Fabiano/05_Projects/03_Studios/53_Edulcowater-Smart/70_Manuals
$ curl -X POST http://192.168.178.130/smartcommand -d "cmd=getdeviceabout"
% Total % Received % Xferd Average Speed Time Time Time
100 177 100 159 100 18 1715 194 --:-- --:-- --:-- 1923{"deviceType":"EducoWater-Premium","hwVersion":"01.00","swVe
rsion":"01.00","deviceId":"E07A04","ssid":"Educo_asdf","password":"123456789","ip":"192.168.4.1"}
DAQUFAB@K123716H1 MINGW64 /c/Fabiano/05_Projects/03_Studios/53_Edulcowater-Smart/70_Manuals
$ |
```

### Implemented Commands

The following pages describe in detail all the GET and SET commands currently implemented in the system.

Each command includes a description, required parameters, limitations, example responses, and a working Bash curl example.

#### GET Commands

-  **getdeviceabout** — Retrieve general device information
-  **getcontrolstatus** — Get the configuration and live status of a control channel
-  **getcalstatus** — Retrieve calibration data and the current sensor value
-  **getsystemstatus** — Get the operating mode, manual status, and system timer configuration

#### SET Commands

-  **setsystemmode** — Set the operating mode to Manual, Timer, Automatic, or SmartHome
-  **setcontroltype** — Set a supported control type for a compatible control channel
-  **setcontrolhyster** — Define the control hysteresis value
-  **setcontrollogik** — Configure direct or inverse control logic
-  **setcontrolsetpoint** — Define the control setpoint
-  **settimeractivation** — Enable or disable a timer block
-  **settontoff** — Set the timer ON and OFF durations
-  **resetcalibration** — Reset a supported sensor calibration point
-  **setsmarthomestatus** — Turn the Smart Home manual output ON or OFF
-  **setsystemtimer** — Set the global system timer ON and OFF durations

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### 1) GET DEVICE ABOUT

#### Description

This command retrieves general information about the EducoWater device. It returns the device identity, firmware and hardware versions, and Access Point network information.

#### Required Parameters

none

#### Limitations

a) None

#### Response

```
{ "deviceType": "EducoWater-Plattform", "hwVersion": "01.00", "swVersion": "01.00", "deviceId": "A1B2C3", "ssid": "EDULCO_1",
  "password": "mypassword", "ip": "192.168.4.1" }
```

#### Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=getdeviceabout"
```

### 2) GET CONTROL STATUS

#### Description

This command retrieves the current status and configuration of a specific control channel. It returns control metadata (type and linked sensor), current measured value, control settings (setpoint, hysteresis, logic), actuator status, and the associated timer status (if used).

#### Required Parameters

ctrl: Control index (must be 0)

#### Limitations

a) Only control 0 is supported

#### Response

```
{ "ctrl": 0, "type": "ph", "active": true, "logic": false, "setpoint": 703, "hysteresis": 25 }
```

#### Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=getcontrolstatus&ctrl=0"
```

### 3) GET CALIBRATION STATUS

#### Description

This command retrieves the calibration data and current measurement value of a specified sensor. It returns the raw calibration points stored in EEPROM, along with the current sensor reading and sensor type.

#### Required Parameters

sensor: Sensor index 0

#### Limitations

a) Only sensor index 0

#### Response

```
{ "sensor": 0, "type": 3, "value": 977, "calX1": 709, "calY1": 100, "calX2": -644, "calY2": 1245 }
```

#### Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=getcalstatus&sensor=0"
```

### 4) GET SYSTEM STATUS

#### Description

This command retrieves the current global operating status of the device. It returns information about the working mode, manual control status, and system timer configuration. This command is useful for remote monitoring and smart home integration.

#### Required Parameters

None.

#### Limitations

None.

#### Response

```
{ "mode": "Automatic", "manual": false, "timer": { "active": true, "timeOn": 300, "timeOff": 120 } }
```

#### Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=getsystemstatus"
```

#### Exception

This command is available only on models that support Manual mode.

### 5) SET SYSTEM MODE (Working Mode)

#### Description

This command sets the global working mode of the device. The system mode defines how the device controls the output:

- Manual // Timer Mode // Automatic Mode // SmartHome

#### Required Parameters

mode : Desired system mode (string, case-sensitive); "Manual", "Timer", "Automatic", "SmartHome"

#### Limitations

- a) mode must be exactly one of: "Manual", "Timer", "Automatic" "SmartHome"
- b) If mode is missing → the command returns an error.

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### ✓ Response

✓ System mode updated to <mode> // ✗ Failed to set system mode

### 💡 Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=setsystemmode&mode=Manual"
```

### ⚠ Exception

This command is available only on models that support Manual mode.

## 6) 🚀 SET CONTROL TYPE

### 📄 Description

This command sets the sensor type associated with a specific control index. Supported types are **pH ORP and Temperature**.

### 📋 Required Parameters

ctrl: Control index 0 // type: Desired type "ph" "orp" "ntc" "ec" "tds" "sal" "diff" "avg"

### ⚠ Limitations

- a) Only control index 0 or 3 supported
- b) Type must be exactly "ph" "orp" "ntc" "ec" "tds" "sal" "diff" "avg"

### ✓ Response

✓ Control type set // ✗ Failed to set control type

### 💡 Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=setcontrolype&ctrl=0&type=ntc"
```

### ⚠ Exception

Available control channels and supported control types depend on the device model. A control type can be assigned only to a channel compatible with the corresponding sensor or input. Refer to the model-specific control compatibility table for the supported channel/type combinations.

## 7) 🌀 SET CONTROL HYSTERESIS

### 📄 Description

This command sets the hysteresis value for a specific control channel. The hysteresis defines the tolerance band around the setpoint to prevent fast toggling of the output.

### 📋 Required Parameters

ctrl: Control index 0 //hysteresis: Value in ADC units (integer)

### ⚠ Limitations

- a) Only control index 0 supported
- b) Hysteresis must be a positive integer

### ✓ Response

✓ Hysteresis set // ✗ Failed to set hysteresis

### 💡 Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=setcontrolhyster&ctrl=0&hysteresis=49"
```

## 8) 🧠 SET CONTROL LOGIC

### 📄 Description

This command sets the logic mode of a control channel. The logic determines how the relay output reacts relative to the setpoint: **true** → Direct logic; **false** → Inverse logic The exact switching behavior depends on the configured control type (pH, ORP, etc.).

### 📋 Required Parameters

ctrl: Control index 0 //logic: "true" or "false"

### ⚠ Limitations

- a) Only control index 0 supported
- b) Logic must be "true" or "false"

### ✓ Response

✓ Logic set // ✗ Failed to set logic

### 💡 Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=setcontrollogik&ctrl=0&logic=false"
```

## 9) 🎯 SET CONTROL SETPOINT

### 📄 Description

This command sets the target setpoint value for a specific control channel.

### 📋 Required Parameters

ctrl: Control index (must be 0) // setpoint: Desired setpoint value (integer)

### ⚠ Limitations

- a) Only control index 0 supported
- b) Setpoint must be a valid integer in ADC range

### ✓ Response

✓ SetPoint set // ✗ Failed to set SetPoint

### 💡 Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=setcontrolsetpoint&ctrl=0&setpoint=704"
```

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## 10) SET TIMER ACTIVATION

### Description

This command activates or deactivates a timer block. Useful for enabling/disabling periodic logic based on time cycles.

### Required Parameters

timer: Timer index 0 // active: "true" or "false"

### Limitations

- a) Timer index must be 0
- b) Active must be "true" or "false"

### Response

✓ Timer activation set // ✗ Failed to set timer activation

### Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=settimeractivation&timer=0&active=false"
```

## 11) SET TIMER TON/TOFF

### Description

This command sets the ON (TON) and OFF (TOFF) durations in seconds for a selected timer. The timer runs cyclically: ON for ton seconds; OFF for toff seconds

### Required Parameters

timer: Timer index 0 // ton: Duration in seconds for ON state // toff: Duration in seconds for OFF state

### Limitations

- a) ton and toff must be between 1 and 64859 seconds
- b) Command accepted only if device is in TIMER mode

### Response

✓ TON and TOFF set // ✗ Failed to set TON or TOFF

### Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=settontoff&timer=0&ton=180&toff=240"
```

## 12) RESET CALIBRATION

### Description

This command resets a specific calibration point of a sensor to its default value. It is typically used to restore factory calibration or to restart the calibration procedure. Each sensor uses two calibration points, identified by the parameter calpoint.

### Required Parameters

sensor: Sensor index 0 // calpoint: 0 → First calibration point; 1 → Second calibration point

### Limitations

- a) Only valid sensor indexes are accepted by the firmware.

### Response

✓ Calibration reset to default // ✗ Reset failed

### Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=resetcalibration&sensor=0&calpoint=1"
```

### Exception

This command does not apply to control types that do not support two-point calibration, such as ntc, temp, diff, and avg. It is available only for sensors and control types that support calibration points.

## 13) SET SMART HOME MODE

### Description

This command sets the device in Smart Home mode. It is used to turn the manual control output ON or OFF from a smart home master like "home Assistant"

### Required Parameters

status : Desired Relay state (string, case-sensitive); "on" → Relay output ON, "off" → Relay output OFF.

### Limitations

- a) status must be exactly "on" or "off".
- b) If status is missing → the command returns an error.

### Response

✓ Manual status updated // { "error": "Missing status parameter" }

### Example (bash curl)

```
curl -X POST http://192.168.178.170/smartcommand -d "cmd=setsmarthomestatus&status=on"
```

### Exception

This command is available only for controls that include and support the Manual control page.

## 14) SET SYSTEM TIMER

### Description

This command sets the global system timer ON and OFF durations. The system timer is used when the device operates in Timer mode, causing the output to switch cyclically between ON and OFF states.

### Required Parameters

timeon : ON duration in seconds (integer) // timeoff : OFF duration in seconds (integer)

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**⚠ Limitations**

- a) Both parameters are required; missing values will return an error.
- b) Valid time ranges depend on the internal firmware logic.

**✅ Response**

- ✅ Timer updated // { "error": "Missing timeon/timeoff parameters" }

**💡 Example (bash curl)**

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=setsystemtimer&timeon=300&timeoff=120"
```

| Document Name       | Rev | Description                         |
|---------------------|-----|-------------------------------------|
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## PLC FUNCTIONS: MODBUS TCP REFERENCE

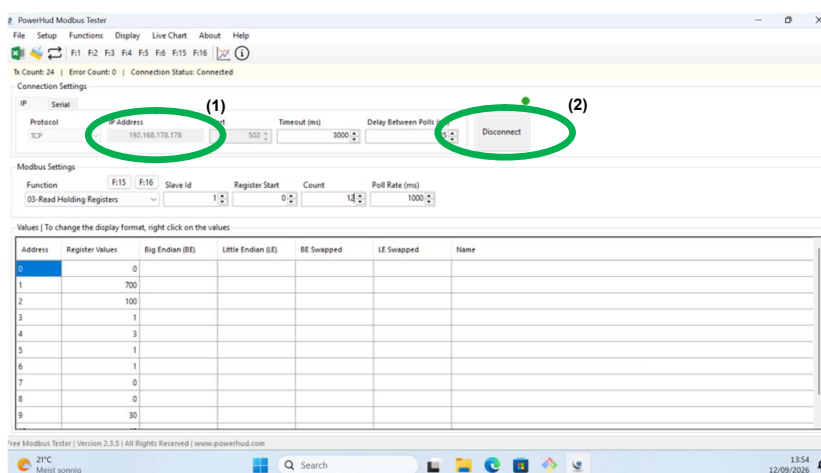
### Using PLC commands

The PLC functions of the EdulcoWater system are designed to enable seamless integration with industrial automation systems, **PLCs**, **HMIs**, **SCADA systems**, and other devices supporting the **Modbus TCP protocol**.

EdulcoWater operates as a **Modbus TCP Server (Slave)**, while the connected PLC or automation controller operates as the **Modbus TCP Client (Master)**. This allows external automation systems, including **Schneider Electric Modicon M221/M241/M262** and **Siemens S7-1200/S7-1500 PLCs**, to read process values and device status and to write control parameters directly through the Ethernet/Wi-Fi network.

The Modbus TCP interface uses **TCP port 502**, **Unit ID / Slave ID 1**, and Holding Registers. **FC03 (Read Holding Registers)** is used for reading; **FC06 (Write Single Register)** for writing one register; and **FC16 / 0x10 (Write Multiple Registers)** for writing multiple consecutive registers.

For development, commissioning, and diagnostic purposes, it is not necessary to have a physical PLC available. The Modbus TCP interface can be tested using any standard Modbus TCP Master/Client testing software, such as the Modbus Tester **ModbusTester 3.2.0** shown in the figure below and available at this link <https://www.modbustester.com/>



Once the Modbus testing software has been installed, launch the application and configure the connection using the IP address of the **EdulcoWater device (1)**

The EdulcoWater device and the **PLC**, or the PC running the PLC/Modbus Master simulator, must be connected to the **same local network/router**.

Enter the EdulcoWater IP address, select **Modbus TCP**, set the communication port to **502** and the **Unit ID to 1**, then press **Connect (2)** to establish the connection with the device.

Once connected, the PLC or Modbus testing software can read and write the parameters exposed by EdulcoWater through the Modbus TCP interface.

### MODBUS TCP REGISTER MAP

The EdulcoWater Industrial Modbus TCP interface provides **60 Holding Registers**, addressed from 0 to 59. The first **48 registers** are divided into **4 control blocks of 12 consecutive registers**. Registers **48–49 are reserved**, and registers **50–59** contain **system parameters**.

**Addressing Note:** The protocol addresses below are zero-based. In 4xxxx notation, address 0 corresponds to 40001, address 1 to 40002, and so on.

The following **Control Registers Table** provides a complete overview of all Modbus control registers available in EdulcoWater, including register addresses, access permissions, data types, units, descriptions, and compatibility with the various EdulcoWater device models

| Document Name       | Rev | Description                          |
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Control Registers Table

| Register | Address | Control | Parameter         | Description                                                     | Access | Device Compatibility                                                 |
|----------|---------|---------|-------------------|-----------------------------------------------------------------|--------|----------------------------------------------------------------------|
| 0        | 40001   | 0       | ActReadVal        | Current value read from the sensor associated with the control. | R      | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 1        | 40002   | 0       | Set_Point         | Control setpoint used by the regulation logic.                  | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 2        | 40003   | 0       | Set_Hyster        | Hysteresis value applied around the configured setpoint.        | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 3        | 40004   | 0       | CtrlSt            | Current control/relay output state. 0 = OFF, 1 = ON.            | R      | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 4        | 40005   | 0       | type              | Type of control/sensor associated with the control channel.     | R      | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 5        | 40006   | 0       | Active            | Enables/disables the control. 0 = Disabled, 1 = Enabled.        | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 6        | 40007   | 0       | Set_PosLogik      | Control logic configuration / operating direction.              | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 7        | 40008   | 0       | HwRelay           | Hardware relay associated with the control. Index 0–3.          | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 8        | 40009   | 0       | Timer.Active      | Enables/disables the control timer. 0 = Disabled, 1 = Enabled.  | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 9        | 40010   | 0       | Timer.OnSeconds   | Timer ON duration in seconds.                                   | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 10       | 40011   | 0       | Timer.OffSeconds  | Timer OFF duration in seconds.                                  | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 11       | 40012   | 0       | Timer.ElapsedTime | Current elapsed timer time.                                     | R      | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 12       | 40013   | 1       | ActReadVal        | Same as for control 0 register 0                                | R      | INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP                     |
| 13       | 40014   | 1       | Set_Point         | Same as for control 0 register 1                                | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP                     |
| 14       | 40015   | 1       | Set_Hyster        | Same as for control 0 register 2                                | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP                     |
| 15       | 40016   | 1       | CtrlSt            | Same as for control 0 register 3                                | R      | INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP                     |
| 16       | 40017   | 1       | type              | Same as for control 0 register 4                                | R      | INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP                     |
| 17       | 40018   | 1       | Active            | Same as for control 0 register 5                                | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP                     |
| 18       | 40019   | 1       | Set_PosLogik      | Same as for control 0 register 6                                | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP                     |
| 19       | 40020   | 1       | HwRelay           | Same as for control 0 register 7                                | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP                     |
| 20       | 40021   | 1       | Timer.Active      | Same as for control 0 register 8                                | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP                     |
| 21       | 40022   | 1       | Timer.OnSeconds   | Same as for control 0 register 9                                | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP                     |
| 22       | 40023   | 1       | Timer.OffSeconds  | Same as for control 0 register 10                               | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP                     |
| 23       | 40024   | 1       | Timer.ElapsedTime | Same as for control 0 register 11                               | R      | INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP                     |
| 24       | 40025   | 2       | ActReadVal        | Same as for control 0 register 0                                | R      | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP |
| 25       | 40026   | 2       | Set_Point         | Same as for control 0 register 1                                | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP |
| 26       | 40027   | 2       | Set_Hyster        | Same as for control 0 register 2                                | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP |
| 27       | 40028   | 2       | CtrlSt            | Same as for control 0 register 3                                | R      | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP |
| 28       | 40029   | 2       | type              | Same as for control 0 register 4                                | R      | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP |
| 29       | 40030   | 2       | Active            | Same as for control 0 register 5                                | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP |
| 30       | 40031   | 2       | Set_PosLogik      | Same as for control 0 register 6                                | R/W    | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP |

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|    |       |    |                              |                                                                   |     |                                                                      |
|----|-------|----|------------------------------|-------------------------------------------------------------------|-----|----------------------------------------------------------------------|
| 31 | 40032 | 2  | HwRelay                      | Same as for control 0 register 7                                  | R/W | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP |
| 32 | 40033 | 2  | Timer.Active                 | Same as for control 0 register 8                                  | R/W | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP |
| 33 | 40034 | 2  | Timer.OnSeconds              | Same as for control 0 register 9                                  | R/W | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP |
| 34 | 40035 | 2  | Timer.OffSeconds             | Same as for control 0 register 10                                 | R/W | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP |
| 35 | 40036 | 2  | Timer.ElapsedTime            | Same as for control 0 register 11                                 | R   | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP |
| 36 | 40037 | 3  | ActReadVal                   | Same as for control 0 register 0                                  | R   | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 37 | 40038 | 3  | Set_Point                    | Same as for control 0 register 1                                  | R/W | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 38 | 40039 | 3  | Set_Hyster                   | Same as for control 0 register 2                                  | R/W | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 39 | 40040 | 3  | CtrlSt                       | Same as for control 0 register 3                                  | R   | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 40 | 40041 | 3  | type                         | Same as for control 0 register 4                                  | R   | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 41 | 40042 | 3  | Active                       | Same as for control 0 register 5                                  | R/W | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 42 | 40043 | 3  | Set_PosLogik                 | Same as for control 0 register 6                                  | R/W | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 43 | 40044 | 3  | HwRelay                      | Same as for control 0 register 7                                  | R/W | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 44 | 40045 | 3  | Timer.Active                 | Same as for control 0 register 8                                  | R/W | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 45 | 40046 | 3  | Timer.OnSeconds              | Same as for control 0 register 9                                  | R/W | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 46 | 40047 | 3  | Timer.OffSeconds             | Same as for control 0 register 10                                 | R/W | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 47 | 40048 | 3  | Timer.ElapsedTime            | Same as for control 0 register 11                                 | R   | INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX     |
| 48 | 40049 | -- | --                           | Reserved for future                                               | --  | ALL                                                                  |
| 49 | 40050 | -- | --                           | Reserved for future                                               | --  | ALL                                                                  |
| 50 | 40051 | -- | REG_SYSTEM_MODE              | System mode: 0 = Manual, 1 = Timer, 2 = Automatic, 3 = SmartHome. | R/W | ALL                                                                  |
| 51 | 40052 | -- | REG_MANUAL_STATUS            | General manual command status. 0 = OFF, 1 = ON.                   | R/W | ALL                                                                  |
| 52 | 40053 | -- | REG_SYSTEM_TIMER_STATUS      | Current system timer status.                                      | R   | ALL                                                                  |
| 53 | 40054 | -- | REG_SYSTEM_TIMER_ON_SECONDS  | System timer ON duration in seconds.                              | R/W | ALL                                                                  |
| 54 | 40055 | -- | REG_SYSTEM_TIMER_OFF_SECONDS | System timer OFF duration in seconds.                             | R/W | ALL                                                                  |
| 55 | 40056 | -- | REG_SYSTEM_TIMER_ELAPSED     | Current elapsed time of the system timer.                         | R   | ALL                                                                  |
| 56 | 40057 | -- | REG_SMARTHOME_STATUS         | Smart Home status. 0 = OFF, 1 = ON.                               | R/W | ALL                                                                  |
| 57 | 40058 | -- | REG_SMARTHOME_CONNECTED      | Smart Home connection status.                                     | R   | ALL                                                                  |
| 58 | 40059 | -- | REG_HW_TYPE                  | EducoWater application hardware type.                             | R   | ALL                                                                  |
| 59 | 40060 | -- | REG_UNIT_ID                  | Modbus Unit ID. Current value: 1.                                 | R   | ALL                                                                  |

### Implemented Commands

A brief explanation and practical examples are also provided for the supported Modbus functions:

- **FC06 – Write Single Register**
- **FC16 – Write Multiple Registers**

These examples can be used both for testing with a Modbus Master simulator and as a reference when configuring the communication with the actual PLC.

### GET Commands

| Document Name       | Rev | Description                         |
|---------------------|-----|-------------------------------------|
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1. **FC03 – Read Holding Registers — Read one or more holding registers**
2. **FC06 – Write Single Register — Write a value to a single holding register.**
3. **FC16 – Write Multiple Registers — Write values to multiple holding registers in a single transaction.**

## 1) **FC03 Read Holding Registers**

### - **Description**

The **FC03 – Read Holding Registers** function is used by the PLC or Modbus TCP Master to read one or more Holding Registers from the EducoWater device as per registers description in table “**Control Registers Table**”

### - **Required Parameters**

EducoWater IP Address: Device local IP address;

TCP Port: **502**

Unit ID: **1**

Function Code: **03** – Read Holding Registers

**Start Address:** First register to be read

**Quantity:** Number of consecutive registers to read

### - **Limitations**

- a) The requested Start Address must correspond to a valid EducoWater Holding Register.
- b) The requested address range must remain within the available Modbus Register Map.
- c) Registers 48 and 49 are reserved for future implementation.
- d) FC03 is a **read-only operation** and cannot be used to modify register values.

### - **Response**

When the request is successfully executed, EducoWater returns the current value of each requested Holding Register.

For example, reading **Control 0** starting from Address 0 with a Quantity of 12 returns the complete Control 0 register block:

*ActReadVal, Set\_Point, Set\_Hyster, CtrlSt, type, Active, Set\_PosLogik, HwRelay, Timer.Active, Timer.OnSeconds, Timer.OffSeconds, and Timer.ElapsedTime.*

## **Example – Modbus TCP Master**

The screenshot shows the PowerHud Modbus Tester interface. The 'Connection Settings' section is configured with IP Address 192.168.178.178, Port 502, and Timeout 3000 ms. The 'Modbus Settings' section shows Function 03-Read Holding Registers, Slave ID 1, Register Start 0, Count 12, and Poll Rate 1000 ms. Below these settings is a table titled 'Values | To change the display format, right click on the values'.

| Address | Register Values | Big Endian (BE) | Little Endian (LE) | BE Swapped | LE Swapped | Name |
|---------|-----------------|-----------------|--------------------|------------|------------|------|
| 0       | 0               |                 |                    |            |            |      |
| 1       | 700             |                 |                    |            |            |      |
| 2       | 100             |                 |                    |            |            |      |
| 3       | 1               |                 |                    |            |            |      |
| 4       | 3               |                 |                    |            |            |      |
| 5       | 1               |                 |                    |            |            |      |
| 6       | 1               |                 |                    |            |            |      |
| 7       | 0               |                 |                    |            |            |      |
| 8       | 0               |                 |                    |            |            |      |
| 9       | 30              |                 |                    |            |            |      |

## 2) **FC06 Write Single Register**

### - **Description**

The **FC06 – Write Single Register** function is used by the PLC or Modbus TCP Master to write a value to a single Holding Register of the EducoWater device as per registers description in table “**Control Registers Table**”.

Only registers identified as **Read/Write (R/W)** can be modified using this function.

### - **Required Parameters**

EducoWater IP Address: Device local IP address;

TCP Port: **502**

Unit ID: **1**

Function Code: **06** – Write Single Register

| Document Name       | Rev | Description                         |
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**Register Address:** Address of the Holding Register to be written

**Value:** New value to be assigned to the selected register

#### - ⚠ Limitations

- The requested Register Address must correspond to a valid EducoWater Holding Register.
- The selected register must be identified as Read/Write (R/W) in the Modbus Register Map.
- Registers identified as Read Only (R) cannot be modified using FC06.
- The written value must comply with the valid range and format defined for the selected register.

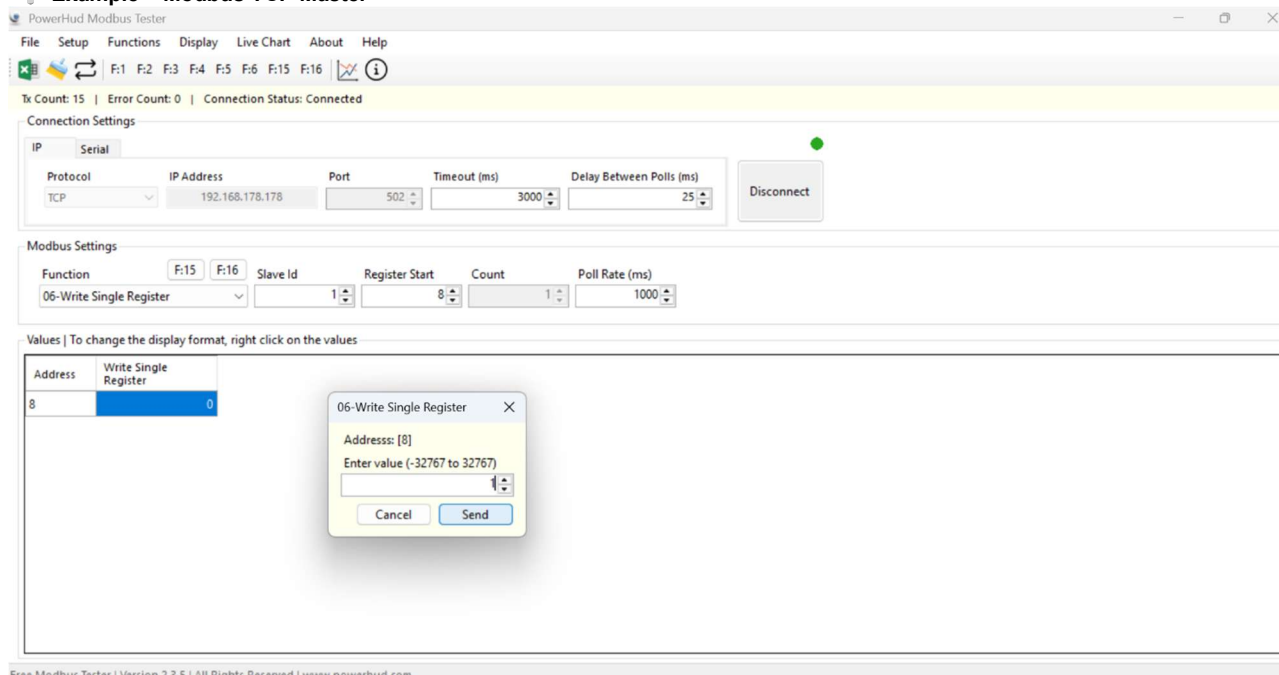
#### - ✅ Response

When the request is successfully executed, EducoWater writes the requested value to the selected Holding Register and returns a confirmation of the operation.

For example, writing to Address 1 modifies the Set\_Point parameter of Control 0.

*The updated value can subsequently be verified by executing an FC03 – Read Holding Registers request on the same register.*

#### - 💡 Example – Modbus TCP Master



### 3) FC16 Write Multiple Registers

#### - 📖 Description

The **FC16 – Write Multiple Registers** function is used by the PLC or Modbus TCP Master to write multiple consecutive Holding Registers of the EducoWater device within a single Modbus TCP request, as per registers description in table “**Control Registers Table**”. Only registers identified as Read/Write (R/W) can be modified.

#### - 📋 Required Parameters

EducoWater IP Address: **Device local IP address**;

TCP Port: **502**

Unit ID: **1**

Function Code: **16 (0x10) – Write Multiple Registers**

**Start Address:** Address of the first Holding Register to be written

**Quantity:** Number of consecutive registers to be written

**Values:** Values to be assigned to the selected consecutive registers

#### - ⚠ Limitations

- The requested Start Address must correspond to a valid EducoWater Holding Register.
- The complete requested address range must remain within the available Modbus Register Map.
- Registers identified as Read Only (R) cannot be modified.
- Each written value must comply with the valid range and format defined for its corresponding register.
- The selected register range must contain only registers that are valid for the requested write operation.

#### - ✅ Response

When the request is successfully executed, EducoWater writes the requested values to the selected consecutive Holding Registers and returns a confirmation containing the starting address and the number of registers written.

For example, FC16 can be used to update multiple consecutive parameters of a Control block within a single Modbus transaction.

*The updated values can subsequently be verified by executing an FC03 – Read Holding Registers request over the same address range.*

| Document Name       | Rev | Description                         |
|---------------------|-----|-------------------------------------|
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## - Example – Modbus TCP Master

PowerHud Modbus Tester

File Setup Functions Display Live Chart About Help

F:1 F:2 F:3 F:4 F:5 F:6 F:15 F:16

Tx Count: 8 | Error Count: 0 | Connection Status: Connected

Connection Settings

IP Serial

Protocol TCP IP Address 192.168.178.178 Port 502 Timeout (ms) 3000 Delay Between Polls (ms) 25

Disconnect

Modbus Settings

Function F:15 F:16 Slave Id Register Start Count Poll Rate (ms)

16-Write Multiple Registers 1 8 2 1000

Values | To change the display format, right click on the values

| Address | Write Multiple Register |
|---------|-------------------------|
| 8       | 0                       |
| 9       | 1                       |

16-Write Multiple Registers

Address: [9]

Enter value (-32767 to 32767)

Cancel Send

Free Modbus Tester | Version 3.2.6 | 68 Dinsley Damarkt | www.powerhud.com

| Document Name       | Rev | Description                         |
|---------------------|-----|-------------------------------------|
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